

“There is a paradigm shift from hardware-centric radios to software-centric radios”

Rohde & Schwarz (R&S) is a test and measurement (T&M) solution provider in the fields of broadcasting, radio monitoring and radio location as well as mission-critical radio communications. For more than 75 years, the company has been developing, producing and marketing a wide range of electronic products.

Yateesh Mohan, managing director, Rohde & Schwarz India, spoke to Dilin Anand of EFY about the trends and challenges while working with the latest technologies

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What are the challenges in designing next-generation T&M products?

The challenge is to have products with a platform capable of absorbing new evolving technologies and form factor suitable for field usage and easy portability. Multiple devices in one, large memory capacity, long battery power, high accuracy, compliance to highest safety standards and finally prices that are substantially lower than the traditional R&D lab equipment are the other requirements.

For R&S, designing products for each of its main verticals requires knowledge in diverse fields encompassing military technology, broadcasting, and also latest telecom technologies like 3G and 4G.

What are the measurement challenges in working with millimetre-wave and terahertz frequencies?

For measurements in microwave and terahertz frequen-

cies, the challenge is to maintain proper environmental lab conditions for complete 24-hour cycle. Supporting accessories and probes and also the local support in providing world-class calibration facilities are the other challenges for these frequency ranges.

To extend the measurement frequency, we use up-converters for signal generation up to 110GHz frequency and down-converters for signal analysis up to 500GHz frequency. The main challenge is to handle the very small connectors and test devices at such frequencies. Also, the accuracy of measurement devices is difficult to predict with a high level of certainty.

Your views on modular instrumentation vs traditional box type?

While PXIe and VXI technologies have been around for more than a decade now, customers still continue to show